higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

16 November 2020

NATED G

This marking guideline consists of 10 pages.

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QUESTION 1: ALTERNATING-CURRENT THEORY

1.1 1.1.1 $T = \frac{L}{R}$

$\therefore L = TxR$

$= 1 \times 10^{-3} \times 10$ ✓

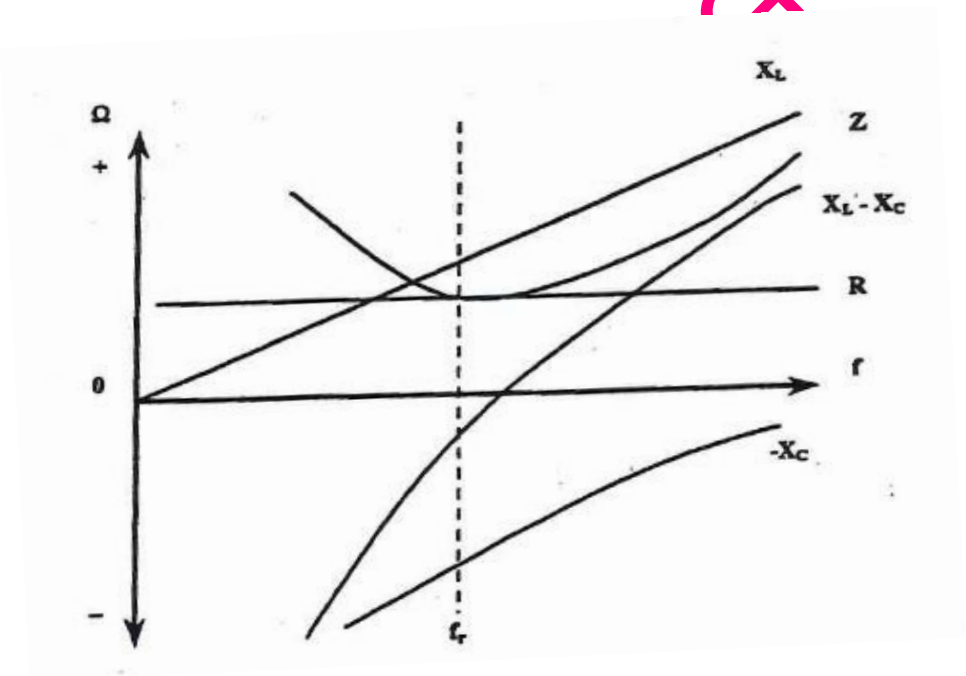
$= 10mA$ ✓ ⟶

1.1.2 $I_L = 0{,}632 \times I_T$

$= 0{,}632 \times 0{,}5$ ✓

$= 0{,}316\ A$ ✓ ⟶ (2 × 2) (4)

1.2



REACTANCE AND IMPEDANCE AS OPPOSING FREQUENCY (3)

1.3 1.3.1

$$f_r = \frac{f_1 + f_2}{2}$$

$$= \frac{415{,}5 \times 10^3 + 384{,}5 \times 10^3}{2} \checkmark$$

$$= 400\ \text{kHz} \leftarrow \checkmark$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$\therefore C = \frac{1}{(2\pi f)^2 L} \checkmark$$

$$= \frac{1}{(2 \times \pi \times 400 \times 10^3)^2 \times 2 \times 10^{-3}} \checkmark$$

$$= 79{,}157\ \text{pf} \checkmark \longrightarrow$$ (5)

1.3.2

$$Q = \frac{1}{10}\sqrt{\frac{L}{C}}$$

$$Q = \frac{1}{10}\sqrt{\frac{2 \times 10^{-3}}{79{,}157 \times 10^{-12}}} \checkmark$$

$$= 251{,}33 \checkmark \longrightarrow$$

OR

$$Q = \frac{X_C}{R}$$

$$= \frac{2 \times \pi \times 400 \times 10^3 \times 2 \times 10^{-3}}{20} \checkmark$$

$$= 251{,}33 \checkmark \longrightarrow$$ (2)

[14]

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QUESTION 2: POWER SUPPLIES

2.1

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL.}}$$

$$0,06 = \frac{20,5 - V_{FL}}{V_{FL}}$$

$$0,06 \times V_{FL} = 20,5 - V_{FL} \checkmark$$

$$V_{FL} = \frac{20,5}{1,06}$$

$$= 19,34\ V \leftarrow \checkmark$$

$$Vd = V_{NL} - V_{FL}$$

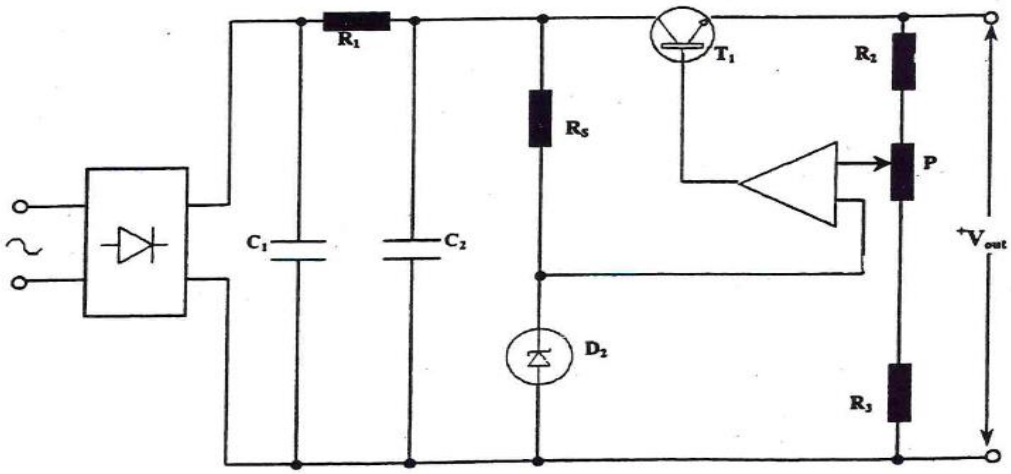
$$= 20,5 - 19,34\ V$$

$$= 1,16\ V \checkmark \longrightarrow$$

(3)

2.2 Capacitor C_1 charges to V_m through diode D_1 during the positive half-cycle. ✓ During the negative half-cycle capacitor C_2 charges to $2V_m$ ✓. During the positive half-cycle diode D_3 conducts and the voltage across capacitor C_2 charges capacitor C_3 to the same $2V_m$ peak voltage. ✓ The voltage across capacitor C_2 is $2V_m$, across C_1 and C_3 it is $3V_m$. ✓ (4)

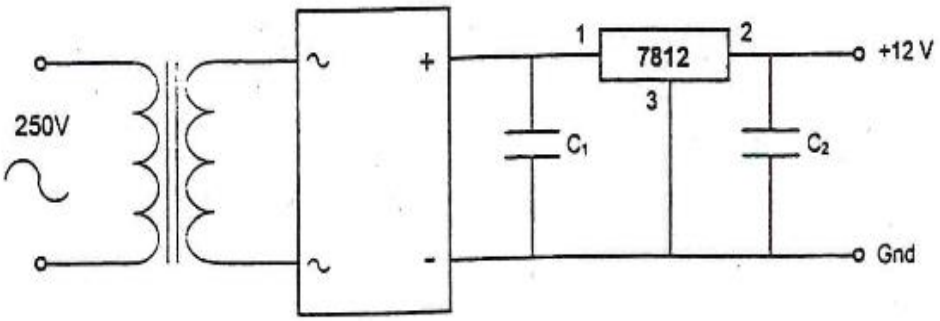
2.3



SERIES VOLTAGE REGULATOR WITH AMPLIFIER (5)

VATED GO

2.4



12 V POWER SUPPLY (4)
[16]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 3.1.1

$$V_e = \frac{1}{10} \times V_{cc}$$

$$= \frac{1}{10} \times 14$$

$$= 1{,}5\ V \leftarrow \checkmark$$

$$R_E = \frac{V_E}{I_E}$$

$$= \frac{1{,}4}{12{,}5 \times 10^{-3}} \checkmark$$

$$= 112\ \Omega \checkmark \longrightarrow$$ (3)

3.1.2

$$Ce = \frac{10}{2\pi f R_e}$$

$$= \frac{10}{2 \times \pi \times 60 \times 112} \checkmark$$

$$= 236{,}84\ \mu f \checkmark \longrightarrow$$ (2)

3.1.3

$$R_C = \frac{V_{cc} - V_{ce} - V_e}{I_C}$$

$$= \frac{14 - 7 - 1{,}4}{12{,}5 \times 10^{-3}} \checkmark$$

$$= 448\ \Omega \checkmark \longrightarrow$$ (2)

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3.1.4 $R_{B2} = \frac{1}{10}\beta R_e$

$= \frac{1}{10} \text{x } 350 \text{ x } 112$ ✓

$= 3920\ \Omega$ ✓ → (2)

3.1.5 $V_B = V_e + V_{be}$

$= 1{,}4 + 0{,}7\ V$

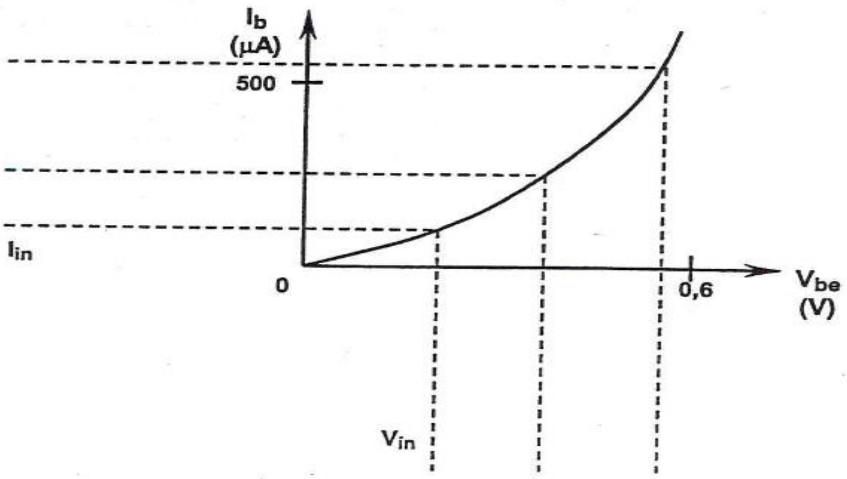
$= 2{,}1\ V \leftarrow$ ✓

$R_{B1} = \frac{R_{B2}(V_{cc} - V_B)}{V_R}$

$= \frac{3920(14 - 2{,}1)}{2{,}1}$ ✓

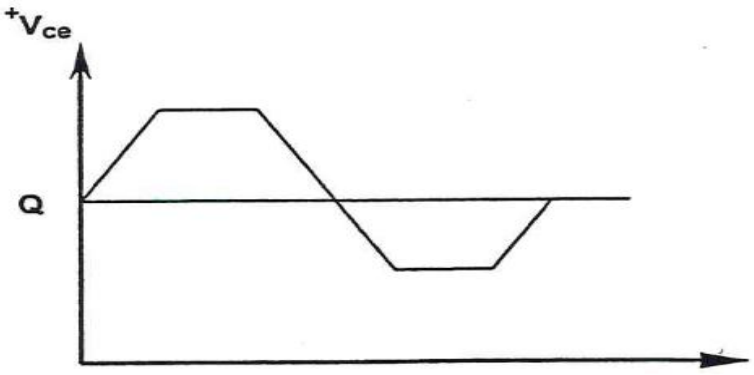
$= 22{,}2133\ k\Omega$ ✓ → (3)

3.2 3.2.1



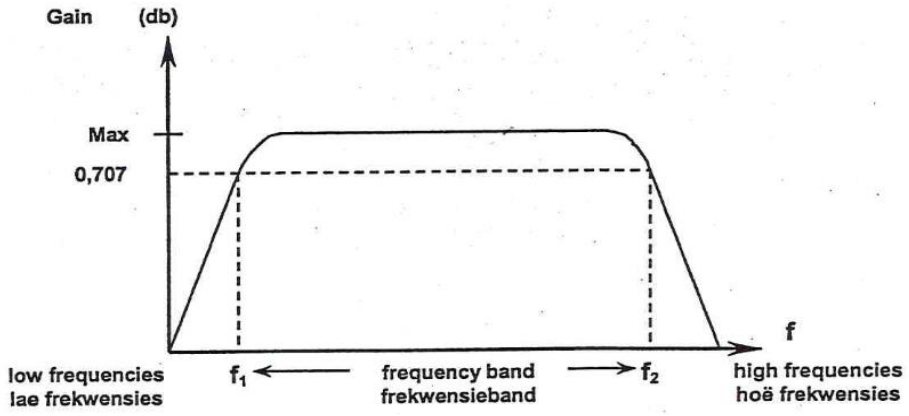
NONLINEAR DISTORTION

3.2.2



AMPLITUDE

3.2.3



FREQUENCY DISTORTION

(3 × 2) (6)
[18]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 The transistors in a differential amplifier circuit show a disorder and cause a small deviation of bias voltage and currents✓ at input terminals which can be amplified effectively by the operational amplifier.✓ The value required at the output is zero.✓ (3)

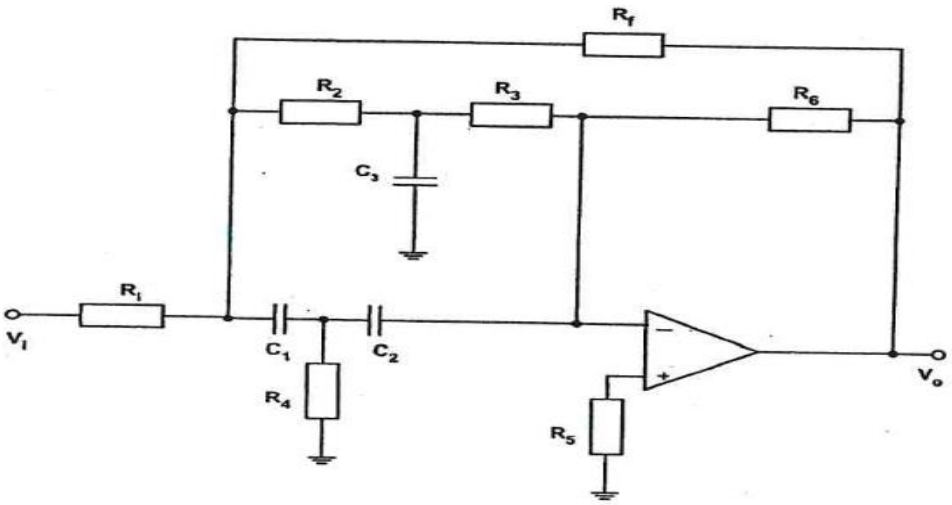
4.2
- Electric analogue computers
- Servo systems
- Linear mixers
- Level shifting

(Any 1 × 1) (1)

4.3
$$V_o = \frac{R_f}{R_1}(V_2 - V_1)$$
$$= \frac{10{,}5\ x\ 10^3}{10{,}5\ x\ 10^3}(3-2)\checkmark\checkmark$$
$$= 1V\checkmark \longrightarrow$$
(3)

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4.4



BAND-REJECTION FILTER (4)
[11]

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QUESTION 5: INTEGRATED CIRCUITS

- Store in conductive sponge
- Store in antistatic containers
- Short-circuit pins with wire
- Push into tin foil on polystyrene (Any 3 × 1) **[3]**

QUESTION 6: TRANSDUCERS

6.1 Passive transducer✓
Needs electrical power to operate✓

Active or self-generating transducer✓
Generates an emf when stimulated by some form of energy and requires no external power✓ (2 + 2) (4)

6.2

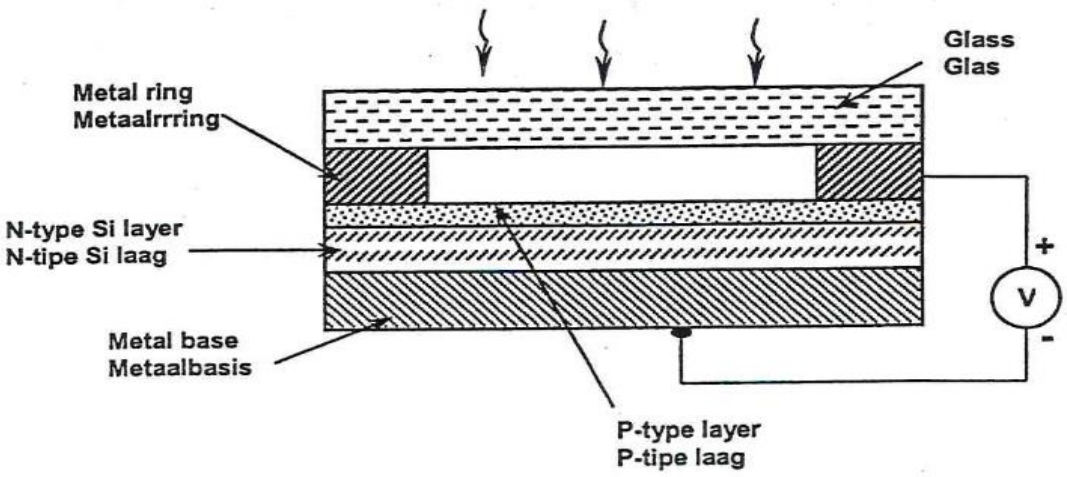
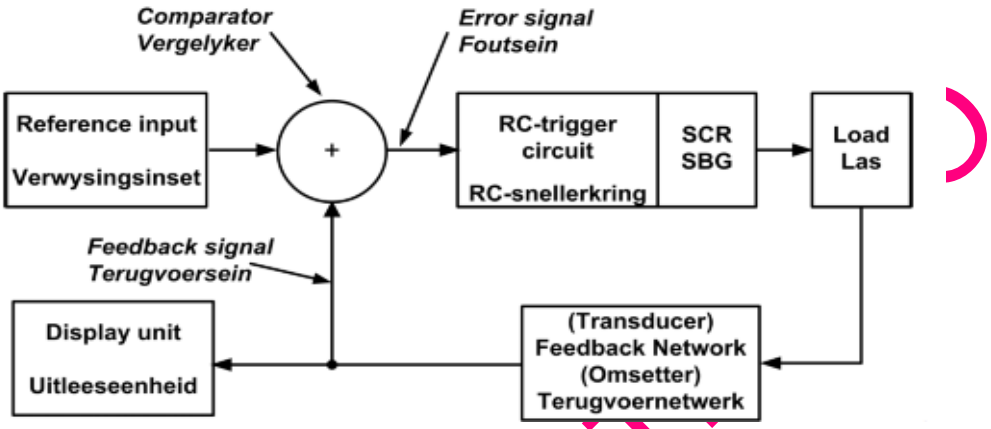


PHOTO-VOLTAIC CELL (5)

6.3 With the core in the centre, equal voltages are indorsed in the secondary windings.✓ Opposing each other, output V_0 is 0V. As external force moves core to left A, ✓ more magnetic flux develops in left winding than right winding (B). Therefore induced voltage in winding A is greater than in winding B.✓Thus size of output voltage equals difference between two secondary voltages of winding A.✓ V_0 is proportional to displacement amount and phase of V_0 is indication of direction in which it moves.✓ (5)
[14]

QUESTION 7: ELECTRONIC PHASE CONTROL



CLOSED-LOOP CIRCUIT (4)

Comparator compares feedback signal from load with reference signal. When load deviates error signal appears on output of comparator, which is used by control circuit to bring load condition back to normal (3)
[7]

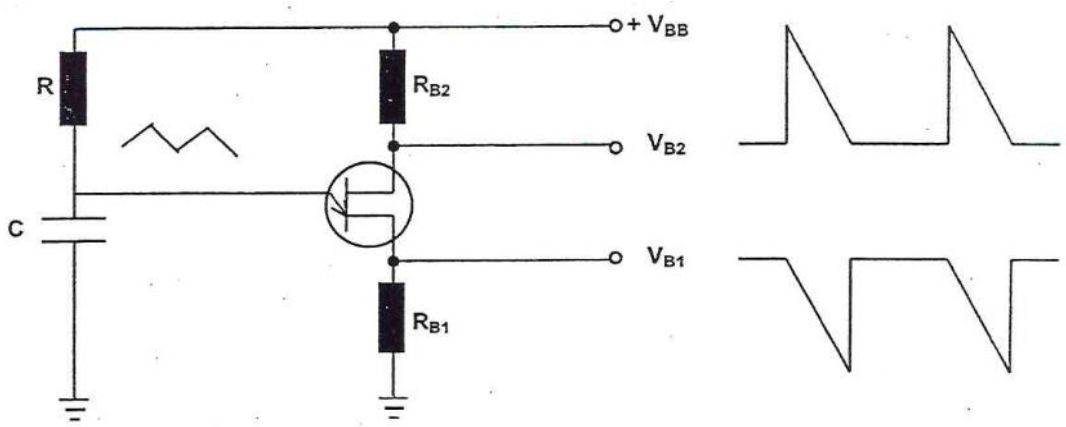
QUESTION 8: TEST EQUIPMENT

Control unit resets circuit to zero. On application of input voltage comparator has an output to clock pulse generator. Clock pulses are passed through to counter, which counts pulses. Digital output is provided to D/A converter, which provides an analogue voltage for comparator. As soon as $V_U = V_r$, comparator output returns to zero, clock pulse generator stops supplying pulses to counter and counter stops counting. Counter serves as indication. **[6]**

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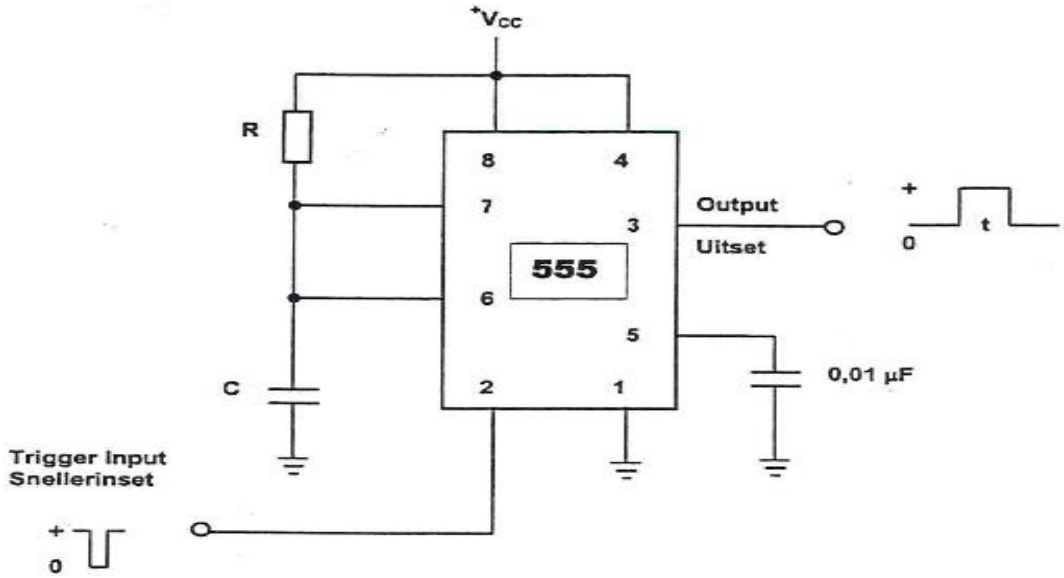
QUESTION 9: OSCLLATORS

9.1



BASIC UJT OSCILLATOR (4)

9.2



555-TIMER IN MONOSTABLE OPERATION (4)

9.3 9.3.1 positive

9.3.2 amplify

9.3.3 parallel

(3 × 1) (3)
[11]

TOTAL: 100

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